Worksheet 2.6

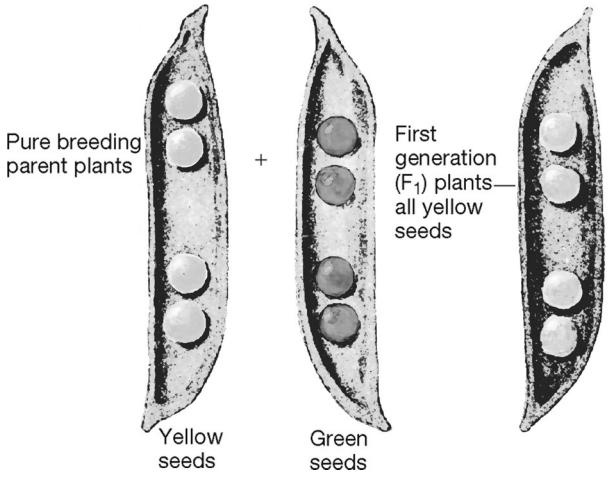
Mendel's experiments

Answers

Green and yellow seeded peas

1. Gregor Mendel experimented with pure breeding yellow-seeded and green-seeded pea plants. He pollinated the flowers of the yellow-seeded variety with pollen from the green-seeded variety.

The results for the first generation (F_1) are shown in the diagram.



(a) Identify which gene is dominant for seed colour. .....Yellow.....

	Y	*Y*
y	*Yy*	*Yy*
y	*Yy*	*Yy*

(b) Use a Punnett square to show that all F_1 plants should be yellow seeded. Use suitable letters for the seed colour characteristic.

All the F_1 are yellow seeded as *Y* is dominant to *y*.

(c) Are the F_1 generation hybrid or pure breeding? .....Hybrids (*Yy*).....

	Y	*y*
Y	*YY*	*Yy*
y	*Yy*	*yy*

(d) The F_1 generation are now interbred. Use a Punnett square to predict the proportion of phenotypes observed in the second generation (F_2).

3:1 ratio of yellow seeded to green seeded

Round and wrinkled seeds

2. Mendel conducted further experiments on pea plants with round seeds or wrinkled seeds. He found that round seeds (*R*) were dominant to wrinkled seeds (*r*).

Complete the Punnett square to predict the proportion of phenotypes when hybrid pea plants are crossed with pure breeding wrinkled seed plants.

	R	*r*
r	*Rr*	*rr*
r	*Rr*	*rr*

1:1 ratio of round to wrinkled

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